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FIRST PRE-BOARD EXAMINATION: 2024-25

SUBJECT: PHYSICS

SET 2

CLASS: XII

TIME: 3Hrs

MAX MARKS: 70

General Instructions:

- (1) There are 33 questions in all. All questions are compulsory.
- (2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E. All these sections are compulsory.
- (3) Section A contains sixteen questions, twelve MCQ and four Assertion- Reasoning based of 1 mark each, Section B contains five questions of two marks each, Section C contains seven questions of three marks each, Section D contains two case study based questions of four marks each and Section E contains three long answer questions of five marks each.
- (4) There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.
- (5) Use of calculators is not allowed.
- (6) You may use the following values of physical constants where ever necessary.

$$\begin{array}{ll} 1. c = 3 \times 10^8 \text{ m/s} & 2. m_e = 9.1 \times 10^{-31} \text{ kg} \\ 3. e = 1.6 \times 10^{-19} \text{ C} & 4. \mu_0 = 4\pi \times 10^{-7} \text{ Tm/A} \\ 5. h = 6.63 \times 10^{-34} \text{ Js} & 6. \epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N m}^2 \end{array}$$

2. 7. Avogadro's number= 6.023×10^{23} gram per mole

SECTION -A

- 1) Current sensitivity of a galvanometer can be increased by decreasing:

(a) Magnetic field B (b) number of turns N (c) Torsional constant K (d) Area A
2. Choose the correct statement.

(a) A capacitor can conduct a dc circuit but not an inductor.
(b) In a dc circuit the inductor can conduct but not a capacitor.
(c) In dc circuit both the inductor and capacitor cannot conduct.
(d) The inductor has infinite resistance in a dc circuit.
3. Electromagnetic waves are produced by
a) constant electric field b) constant magnetic field
c) changing electric and magnetic field d) constant electric and magnetic field
4. A ray of light travels from glass (refractive index=1.5) to air (refractive index=1). If the critical

angle is 42° , the angle of deviation for an angle of incidence of 50° will be:

- a) 60.0° b) 22.0° c) 45.0° d) 80.0°

5. In the phenomena of Diffraction of light when the violet light is used in the experiment instead of red light then,

- a) Fringe width increases b) No change in fringe width
c) Fringe width decreases d) Colour pattern is formed

6. The work function of a metal is independent of

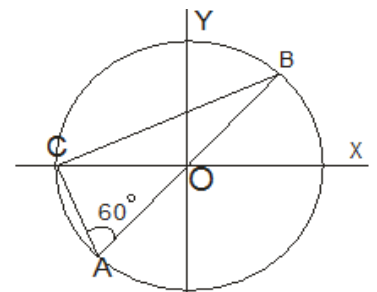
- (i) nature of the surface of the metal (ii) dimensions of the metal
(iii) properties of the metal (iv) abundance of the metal
(a) (i) only (b) (i) and (iii) (c) (ii) and (iii) (d) (ii) and (iv)

7. The Balmer series for the H-atom can be observed

- (a) if we measure the frequencies of light emitted when an excited atom falls to the ground state.
(b) if we measure the frequencies of light emitted due to transitions between excited states and the first excited state.
(c) in any transition in a H-atom.
(d) as a sequence of frequencies with the lower frequencies getting closely packed.

8. Consider a system of three charges $q/3$, $q/3$ and $-2q/3$ placed at points A, B and C respectively as shown in the figure. Take O to be the centre of the circle of radius R and angle CAB = 60°

- (a) The electric field at point A is $q^2/8\pi\epsilon_0 R^2$
(b) The magnitude of the force between the charges at C and B is $q^2/54\pi\epsilon_0 R^2$
(c) The potential energy of the system is zero
(d) The electric field at point B is $q^2/12\pi\epsilon_0 R$



9. A conductor of $10\ \Omega$ is connected across a 6 V ideal source. The power supplied by the source to the conductor is:

- (a) 1.8 W (b) 2.4 W (c) 3.6 W (d) 7.2 W

10. In a LCR-circuit, the potential difference between the terminals of the inductance is 60 V, between terminals of the capacitor is 30 V and that between the terminals of the resistance is 40 V. The supply voltage will be

- (a) 25 V (b) 50 V (c) 100 V (d) 200 V

- 11 . The Bohr model for the spectra of a H-atom
- will be applicable to hydrogen in the molecular form.
 - will not be applicable as it is for a He-atom.
 - is valid only at room temperature.
 - predicts continuous as well as discrete spectral lines.
12. Electromagnetic waves with wavelength λ are used by a FM radio station for broadcasting. Here λ belongs to
- radio waves
 - VHF radio waves
 - UHF radio waves
 - microwaves

For Question 13 to 16, two statements are given – one labeled as Assertion (A) and other labelled Reason(R). Select the correct answer to these questions from the options as given below:

- Assertion and Reason are true and Reason is the correct explanation of Assertion.
 - Assertion and Reason are true but Reason is Not the correct explanation of Assertion.
 - Assertion is true but Reason is false.
 - Both Assertion and Reason are false.
13. **Assertion(A):** The resistance of a given mass of copper wire is inversely proportional to the square of length.
Reason(R): When a copper wire of a given mass is stressed to increase its length, its cross-sectional area also increases.
14. **Assertion(A):** Microwaves are considered suitable for radar systems.
Reason(R): Microwaves are of shorter wavelength.
15. **Assertion(A):** A charge moving in magnetic field experiences a force.
Reason (R): Magnetic field is always associated with moving charge which interacts with the external magnetic field.
16. **Assertion (A):** The coils of a spring come close to each other, when current is passed through it.
Reason (R): It is because, the coils of a spring carry current in the same direction and hence attract each other.

SECTION – B

- Draw the circuit diagram of a Wheat Stone bridge. Obtain the condition when no current flows through the galvanometer in it.
- Write Einstein's photoelectric equation and point out any two characteristic properties of photons on which this equation is based.
- Both, the electric and magnetic fields can deflect a moving electron. What is the difference between these deflections?
- Derive an expression for the radius of nth Bohr's orbit in Hydrogen atom.

OR

Energy of electron in first excited state in Hydrogen atom is -3.4eV . Find KE and PE of electron in the ground state

21. The ratio of maximum and minimum intensities of two sources is 4:1. Find the ratio of their amplitudes.

SECTION – C

22. Dhruv took three capacitors namely C_1 , C_2 & C_3 having capacitances $3\mu\text{F}$, $4\mu\text{F}$ and $2\mu\text{F}$ and placed C_1 in series with the parallel combination of C_2 & C_3 . He applied a potential difference V_{AB} of +1200V across C_1 and grounded the end point C of the parallel combination.

- (i) Draw the required circuit diagram.
- (ii) Evaluate the charge on each capacitor
- (iii) Find the potential at B

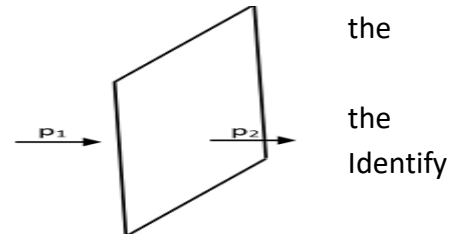
23. (I) A uniform magnetic field gets modified as shown below when two specimens X and Y are placed in it. Identify whether specimens X and Y are diamagnetic, paramagnetic or ferromagnetic



- (ii) How is the magnetic permeability of specimen X different from that of specimen Y?

24. Draw a ray diagram showing the image formation by a compound microscope. Hence obtain expression for total magnification when the image is formed at infinity.

- 25 (a) An electric dipole is kept first to the left and then to right of a negatively charged infinite plane sheet having a uniform surface charge density. The arrows p_1 and p_2 show directions of its electric dipole moment in the two cases. for each case, whether the dipole is in stable or unstable equilibrium. Justify each



- (b) Next, the dipole is kept in a similar way (as shown), near an infinitely long straight wire having uniform negative linear charge density. Will the dipole be in equilibrium at these two positions? Justify your answer.



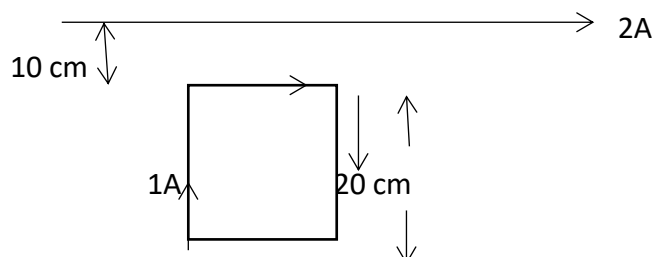
26. A resistor of 30Ω and a capacitor of $250/\pi\mu\text{F}$ are connected in series to a 200V, 50Hz ac source. Calculate (i) the current in the circuit, and (ii) voltage drops across the resistor and the capacitor. (iii) Is the algebraic sum of these voltages more than the source voltage? If yes, solve the paradox.

(OR)

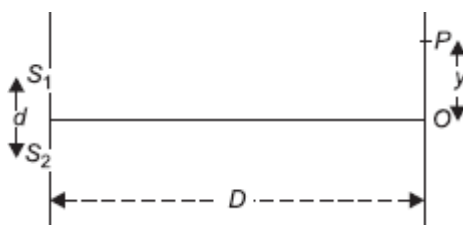
A series LCR circuit with $R=20\Omega$, $L=2\text{H}$ and $C=50\text{F}$ are connected to a 200 volts ac source of variable frequency. What is

- (i) the amplitude of the current,
- (ii) the average power transferred to the circuit in one complete cycle, at resonance?
- (iii) Calculate the potential drop across the capacitor.

27. A square loop of side 20 cm carrying current of 1 A kept near an infinite long straight wire carrying a current of 2 A in the same plane as shown in the figure. Calculate the magnitude and direction of the net force exerted on the loop due to the current carrying conductor.



28. The intensity at the central maxima (O) in a Young's double slit experiment is I_0 . If the distance OP equals one-third of the fringe width of the pattern, show that the intensity at point P would be $I_0/4$.



SECTION – D

29 . In 1905, Albert Einstein explained the photoelectric effect, a phenomenon observed when light strikes a metal surface and ejects electrons. According to Einstein, light behaves as particles called photons, each having an energy of $h\nu$, where h is Planck's constant and ν is the frequency of light. For electrons to be ejected from the surface of the metal, the energy of the photons must exceed a certain minimum value known as the work function (Φ) of the metal. If the frequency of the incident light is less than the threshold frequency (ν_0), no photoelectric emission occurs regardless of the intensity of the light. This challenged the classical wave theory of light, which could not explain why the emission depended on frequency rather than intensity.

1. Which of the following parameters determines whether photoelectric emission occurs?

- | | |
|------------------------|------------------------|
| a) Intensity of light | b) Threshold frequency |
| c) Wavelength of light | d) Angle of incidence |

2. If the frequency of light is increased while keeping the intensity constant, which of the following increases in the photoelectric effect?

- | | |
|--------------------------------|--------------------------------|
| a) Number of emitted electrons | b) Energy of emitted electrons |
| b) Work function | d) Threshold frequency |

3. What does the photoelectric effect demonstrate about the nature of light?

- | | |
|---------------------|-------------------------|
| a) Wave-like nature | b) Particle-like nature |
|---------------------|-------------------------|

b) Dual nature

d) Corpuscular theory

4. Which law of classical physics was contradicted by the results of the photoelectric effect experiment?

a) Newton's second law

b) Maxwell's equations.

c) Law of conservation of energy

d) Wave theory of light

(OR)

4. If the frequency of incident light is doubled, how is the energy of the photons affected?

a) Doubled

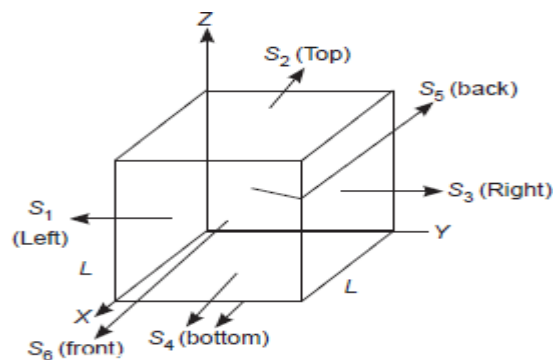
b) Halved

c) Quadrupled

D) Unchanged

30. In electrostatics, electric flux is the measure of the electric field through a given surface, although an electric field in itself cannot flow. It is a way of describing the electric field strength at any distance from the charge causing the field. Now, consider a cube of each edge 0.30 m is placed with its one corner at the origin. The cube is placed in a non-uniform electric field.

$$\vec{E} = (-2x\hat{i} + 3\hat{j}) \text{ N/C}$$



(i) The surfaces that have zero electric flux are

(a) S_1 and S_2

(b) S_1 and S_6

(c) S_2 and S_4

(d) S_1 and S_3

(ii) Electric flux passing through surface S_1 is

(a) $-0.27 \text{ Nm}^2\text{C}^{-1}$

(b) $0.27 \text{ Nm}^2\text{C}^{-1}$

(c) $-0.18 \text{ Nm}^2\text{C}^{-1}$

(d) $0.18 \text{ Nm}^2\text{C}^{-1}$

(iii) Electric flux passing through surface S_4 is

(a) $-0.18 \text{ Nm}^2\text{C}^{-1}$

(b) $+0.18 \text{ Nm}^2\text{C}^{-1}$

(c) $+0.27 \text{ Nm}^2\text{C}^{-1}$

(d) zero

(iv) Total net flux passing through the cube if $\vec{E} = 2\hat{i} \text{ N/C}$

(a) Zero

(b) $-0.18 \text{ Nm}^2\text{C}^{-1}$

(c) $0.18 \text{ Nm}^2\text{C}^{-1}$

(d) $0.27 \text{ Nm}^2\text{C}^{-1}$

(OR)

(v) Total charge enclosed inside the cube is

- (a) 0 (b) -1.62 pC (c) +1.62 pC (d) 2.4 pC

SECTION –E

31. i) What is the principle of a capacitor?

ii) Derive the equation for capacitance of a parallel plate capacitor with a dielectric slab

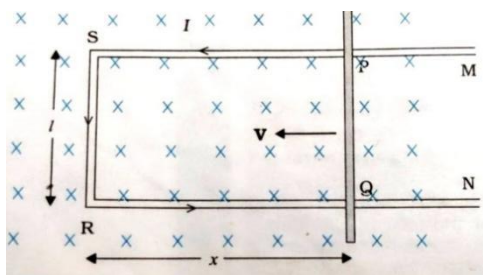
iii) What type of energy is stored between the two plates of a charged capacitor?

(OR)

i) Derive the equation for potential at any point due to a short dipole.

ii) Mention any two properties of equipotential surface.

32 A straight conductor PQ of resistance R is moving in a uniform and time dependent magnetic field as shown in the given figure.



Find:

a.(i) the emf induced across the rod PQ.

(ii) force on the movable arm PQ.

(iii) The power delivered by the external force to maintain the motion of movable arm PQ. 3M

b. In a coil of area 20 cm^2 and 10 turns with magnetic field directed perpendicular to the plane changing at the rate of 10^3 T/s . The resistance of the coil is 20 ohm. What will be the current of the coil? 2M

(OR)

(a) Calculate self-inductance for a long solenoid of length l , number of turns N and radius r .

(b) A coil is mechanically rotated with constant angular speed ω in a uniform magnetic field which is perpendicular to the axis of rotation of the coil. The plane of the coil is initially held perpendicular to the field. Plot a graph showing variation of (i) magnetic flux (Φ) and (ii) the induced emf in the coil as a function of ωt .

33. A thin convex lens having two surfaces of radii of curvature R_1 and R_2 is made of a material of refractive index μ_2 . It is kept in a medium of refractive index μ_1 . Derive, with the help of a ray diagram,

the lens maker's formula when a point object placed on the principal axis in front of the radius of curvature R_1 produces an image I on the other side of the lens.

OR

Draw a labeled ray diagram of an astronomical telescope for the near point adjustment.

You are given three lenses of powers 0.5 D, 4 D, 10 D. State, with reason, which two lenses will you select for constructing a good astronomical telescope.

Derive the expression for magnifying power when the final image is at infinity.